Problem Set 2

Grant Talbert
February 5, 2025
PY 451 Section A1

Boston University

Problem 1

i feel bad for whoever's grading this

1. $$\langle b_1|A|a_1\rangle = \begin{pmatrix}-1 & -i\end{pmatrix}^* \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} \begin{pmatrix}3i \\ 2\end{pmatrix} = \begin{pmatrix}-1 & i\end{pmatrix} \begin{pmatrix}3i+4i \\ -6i+2i\end{pmatrix} = -7i+4.$$

2. $$\langle b_1|A^\dagger|a_1\rangle = \begin{pmatrix}-1 & -i\end{pmatrix}^* \begin{pmatrix}1 & -2 \\ 2i & i\end{pmatrix}^* \begin{pmatrix}3i \\ 2\end{pmatrix} = \begin{pmatrix}-1 & i\end{pmatrix} \begin{pmatrix}1 & -2 \\ -2i & -i\end{pmatrix} \begin{pmatrix}3i \\ 2\end{pmatrix} = \begin{pmatrix}-1 & i\end{pmatrix} \begin{pmatrix}3i-4 \\ -6-2i\end{pmatrix}$$
$$= (4-3i) + (2+6i) = 6+3i.$$

3. $$(\langle b_1|A|a_1\rangle)^\dagger = (-7i+4)^* = 4+7i.$$

4. $$A|a_1\rangle = \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} \begin{pmatrix}3i \\ 2\end{pmatrix} = 3i \begin{pmatrix}1 \\ -2\end{pmatrix} + 2 \begin{pmatrix}2i \\ i\end{pmatrix} = \begin{pmatrix}7i \\ -4i\end{pmatrix}.$$

5. $$\langle b_1|A = \begin{pmatrix}-1 & -i\end{pmatrix}^* \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} = \begin{pmatrix}-1 & i\end{pmatrix} \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} = \begin{pmatrix}-1-2i & -2i-1\end{pmatrix}.$$

6. $$\langle b_1 \mid a_1\rangle = \begin{pmatrix}-1 & -i\end{pmatrix}^* \begin{pmatrix}3i \\ 2\end{pmatrix} = \begin{pmatrix}-1 & i\end{pmatrix} \begin{pmatrix}3i \\ 2\end{pmatrix} = -3i+2i = -i.$$

7. $$\langle a_1 \mid b_1\rangle = (\langle b_1 \mid a_1\rangle)^* = (-i)^* = i.$$

8. $$|b_1\rangle\langle a_1| = \begin{pmatrix}-1 \\ -i\end{pmatrix} \begin{pmatrix}3i & 2\end{pmatrix}^* = \begin{pmatrix}-1 \\ -i\end{pmatrix} \begin{pmatrix}-3i & 2\end{pmatrix} = \begin{pmatrix}3i & -2 \\ -3 & -2i\end{pmatrix}.$$

9. $$A|b_1\rangle\langle a_1| = \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} \begin{pmatrix}3i & -2 \\ -3 & -2i\end{pmatrix} = \begin{pmatrix}3i-6i & -2+4 \\ -6i-3i & 4+2\end{pmatrix} = \begin{pmatrix}-3i & 2 \\ -9i & 6\end{pmatrix}.$$

10. $$A|b_1\rangle\langle a_1|A = \begin{pmatrix}-3i & 2 \\ -9i & 6\end{pmatrix} \begin{pmatrix}1 & 2i \\ -2 & i\end{pmatrix} = \begin{pmatrix}-3i-4 & 6+2i \\ -9i-12 & 18+6i\end{pmatrix}.$$

11. why did we have to do this

$$A^2|b_1\rangle\langle a_1| = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -3i & 2 \\ -9i & 6 \end{pmatrix} = \begin{pmatrix} -3i - 18 & 2 + 12i \\ 6i + 9 & -4 + 6i \end{pmatrix}.$$

12.

$$(|b_1\rangle\langle a_1|)^2 = \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix} \begin{pmatrix} 3i & -2 \\ -3 & -2i \end{pmatrix} = \begin{pmatrix} -9+6 & -6+4i \\ -9i+6i & 6-4 \end{pmatrix} = \begin{pmatrix} -3 & -2i \\ -3i & 2 \end{pmatrix}.$$

13.

$$A^\dagger|a_1\rangle = \begin{pmatrix} 1 & -2 \\ 2i & i \end{pmatrix}^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -2i & -i \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} = \begin{pmatrix} 3i - 4 \\ 6 - 2i \end{pmatrix}.$$

14. why do $|a_1\rangle$ and $|b_1\rangle$ have subscripts if we're only using a single a and b like we could just do $|a\rangle$ and $|b\rangle$ and it would be the same

$$(A|b_1\rangle)^\dagger = \left(\begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^\dagger = \begin{pmatrix} -1+2 \\ 2+1 \end{pmatrix}^\dagger = \begin{pmatrix} 1 & 3 \end{pmatrix}.$$

15.

$$A|b_1\rangle - \lambda|b_1\rangle = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} -1 \\ -i \end{pmatrix} - (2+3i) \begin{pmatrix} -1 \\ -i \end{pmatrix} = \begin{pmatrix} -1+2 \\ 2+1 \end{pmatrix} - \begin{pmatrix} -2-3i \\ -2i+3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2+3i \\ -3+2i \end{pmatrix} = \begin{pmatrix} 5+3i2i \end{pmatrix}.$$

16.

$$\lambda A = (2+3i) \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = \begin{pmatrix} 2+3i & 4i-6 \\ -4-6i & 2i-3 \end{pmatrix}.$$

17.

$$(\lambda - \langle a_1 \,|\, b_1\rangle)A = (2+3i-i)A = (2+2i) \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} = 4i - 4 \begin{pmatrix} 2+2i & \\ -4-4i & 2i-2 \end{pmatrix}.$$

18.

$$(\langle a_1 \,|\, a_1\rangle + \langle b_1 \,|\, b_1\rangle)^{-1} = \left(\begin{pmatrix} 3i & 2 \end{pmatrix}^* \begin{pmatrix} 3i \\ 2 \end{pmatrix} + \begin{pmatrix} -1 & -i \end{pmatrix}^* \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^{-1}$$

$$= \left(\begin{pmatrix} -3i & 2 \end{pmatrix} \begin{pmatrix} 3i \\ 2 \end{pmatrix} + \begin{pmatrix} -1 & i \end{pmatrix} \begin{pmatrix} -1 \\ -i \end{pmatrix} \right)^{-1} = (9+4+1+1)^{-1} = \frac{1}{15}.$$

19.

$$(|a_1\rangle\langle a_1| + |b_1\rangle\langle b_1|)^{-1} = \left(\begin{pmatrix} 3i \\ 2 \end{pmatrix} \begin{pmatrix} -3i & 2 \end{pmatrix} + \begin{pmatrix} -1 \\ -i \end{pmatrix} \begin{pmatrix} -1 & i \end{pmatrix} \right)^{-1} = \left(\begin{pmatrix} 9 & 6i \\ -6i & 4 \end{pmatrix} + \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \right)^{-1}$$

$$= \begin{pmatrix} 10 & 5i \\ -5i & 5 \end{pmatrix}^{-1} = \frac{1}{50+25} \begin{pmatrix} 5 & -5i \\ 5i & 10 \end{pmatrix} = \frac{1}{15} \begin{pmatrix} 1 & -i \\ i & 2 \end{pmatrix}.$$

20.

$$AA^\dagger = \begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2i & -i \end{pmatrix} = \begin{pmatrix} 1+4 & -2+2 \\ -2+2 & 4+1 \end{pmatrix} = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}.$$

21.

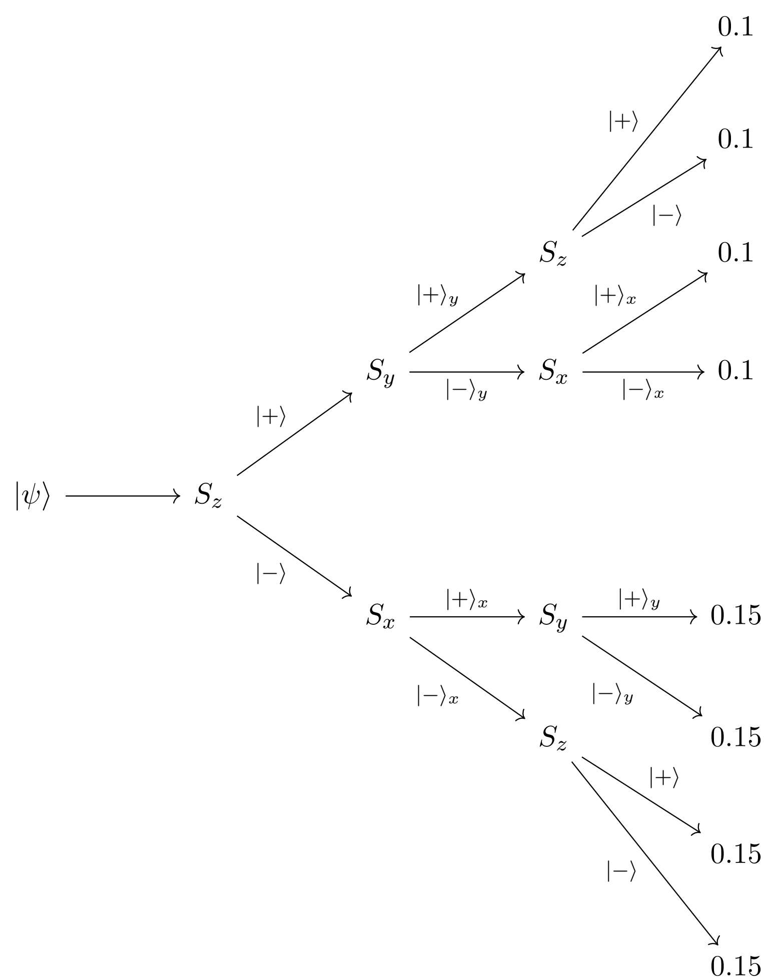
$$\left(A^2\right)^\dagger = \left(\begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix}\begin{pmatrix} 1 & 2i \\ -2 & i \end{pmatrix}\right)^\dagger = \begin{pmatrix} 1-4i & 2i-2 \\ -2-2i & -4i-1 \end{pmatrix}^\dagger = \begin{pmatrix} 1+4i & -2+2i \\ -2-2i & -1+4i \end{pmatrix}.$$

22.

$$\left(AA^\dagger\right)^\dagger = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}^\dagger = \begin{pmatrix} 5 & \\ & 5 \end{pmatrix}.$$

Problem 2

1. Since we know the state after the first measurement, we should know all subsequent probabilities. We have shown multiple times that measurements of a basis state in an orthogonal basis yield probabilities of $1/2$ for each state, so we have that every measurement that originated at $S_z \to |+\rangle$ should be the same. Using this fact, we find that the measurements for $S_z \to |-\rangle$ have 0.15 probability, to ensure the sum of the probabilities is 1.



2. First we find the norm of the given vector.

$$3|+\rangle_x + 5i|-\rangle_x + 7|+\rangle_y \cong \frac{3}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{5i}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + \frac{7}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix} =: \frac{1}{\sqrt{2}}\begin{pmatrix} 10+5i \\ 3+2i \end{pmatrix}\vec{v}.$$

$$\|\vec{v}\| = \frac{1}{\sqrt{2}}\sqrt{|10+5i|^2 + |10+5i|^2} = \frac{1}{\sqrt{2}}\sqrt{125+15} = \frac{\sqrt{140}}{\sqrt{2}} = \sqrt{70}.$$

It follows that

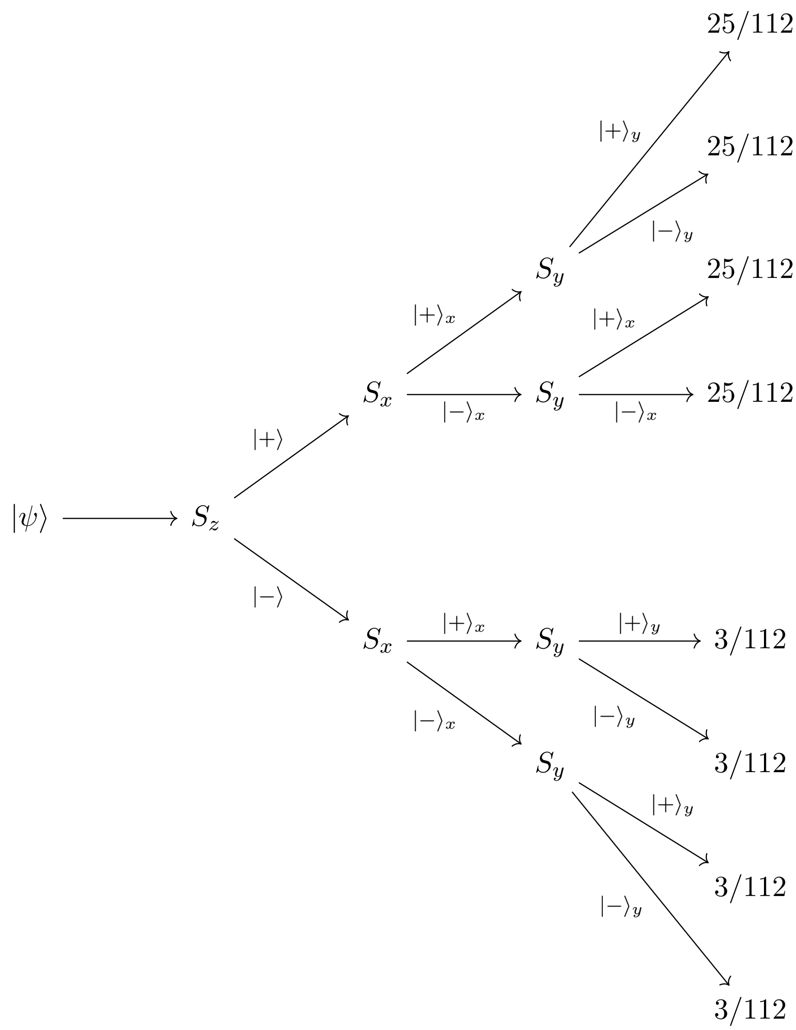
$$|\psi\rangle = \frac{1}{\sqrt{70}}\left(3|+\rangle_x + 5i|-\rangle_x + 7|+\rangle_y\right).$$

3. First, we compute the probabilities after the Z Stern-Gerlach machine.

$$\begin{aligned}
\mathcal{P}_+ &= |\langle +|S_z|\psi\rangle|^2 \\
&= \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}1 & \\ & -1\end{pmatrix}\left(\frac{1}{\sqrt{70}}\left(3|+\rangle_x + 5i|-\rangle_x + 7|+\rangle_y\right)\right)\right|^2 \\
&= \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}1 & \\ & -1\end{pmatrix}\frac{1}{\sqrt{140}}\begin{pmatrix}10+5i \\ 3+2i\end{pmatrix}\right|^2 \\
&= \left|\frac{1}{\sqrt{140}}\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}10+5i \\ -3-2i\end{pmatrix}\right|^2 \\
&= \frac{1}{140}|10+5i|^2 \\
&= \frac{125}{140} \\
&= \frac{25}{28} \\
\mathcal{P}_- &= |\langle -|S_z|\psi\rangle|^2 \\
&= \frac{1}{140}|3+2i|^2 \\
&= \frac{15}{140} \\
&= \frac{3}{28}.
\end{aligned}$$

After the first measurement, since we are measuring every time the particle goes through a machine, we will necessarily be in a basis state, so the distribution afterwards will be even.

We simply divide our output by the number of possibe outcomes to obtain each probability.



Problem 3

Given a unit vector of the form $\hat{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, the spin matrix $S_{\hat{n}}$ is given as

$$S_{\hat{n}} = \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}.$$

In the problem, $\vec{n} = (\sin\theta, 0, \cos\theta)$ is of this form given that $\phi = 0$. We thus have

$$S_{\vec{n}} = \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

We also have

$$\begin{aligned}
S_{\vec{n}}|\psi_1\rangle &\cong \frac{\hbar}{2}\begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}\begin{pmatrix}\cos(\theta/2)\\ \sin(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\cos\theta\cos(\theta/2)+\sin\theta\sin(\theta/2)\\ \sin\theta\cos(\theta/2)-\cos\theta\sin(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\cos\theta\cos(\theta/2)+2\sin^2(\theta/2)\cos(\theta/2)\\ 2\sin(\theta/2)\cos^2(\theta/2)-\cos\theta\sin(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\left(2\cos^2(\theta/2)-1\right)\cos(\theta/2)+2\sin^2(\theta/2)\cos(\theta/2)\\ 2\sin(\theta/2)\cos^2(\theta/2)-\left(2\cos^2(\theta/2)-1\right)\sin(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}2\left(\cos^2(\theta/2)+\sin^2(\theta/2)\right)\cos(\theta/2)-\cos(\theta/2)\\ 2\sin(\theta/2)\cos^2(\theta/2)-\left(2\cos^2(\theta/2)\sin(\theta/2)-\sin(\theta/2)\right)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\cos(\theta/2)\\ 2\left(\cos^2(\theta/2)-\cos^2(\theta/2)\right)\sin(\theta/2)+\sin(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\cos(\theta/2)\\ \sin(\theta/2)\end{pmatrix}\\
&\cong \frac{\hbar}{2}|\psi_1\rangle.
\end{aligned}$$

$$\begin{aligned}
S_{\vec{n}}|\psi_2\rangle &\cong \frac{\hbar}{2}\begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}\begin{pmatrix}\sin(\theta/2)\\ -\cos(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}\cos\theta\sin(\theta/2)-\sin\theta\cos(\theta/2)\\ \sin\theta\sin(\theta/2)+\cos\theta\cos(\theta/2)\end{pmatrix}\\
&= \begin{pmatrix}-(\sin\theta\cos(\theta/2)-\cos\theta\sin(\theta/2)\\ \sin\theta\sin(\theta/2)+\cos\theta\cos(\theta/2)\end{pmatrix}\\
&= \frac{\hbar}{2}\begin{pmatrix}-\sin(\theta/2)\\ \cos(\theta/2)\end{pmatrix}\\
&\cong -\frac{\hbar}{2}|\psi_2\rangle.
\end{aligned}$$

Here I've used the fact that the top entry of $S_{\vec{n}}|\psi_2\rangle$ is the negative of the bottom entry of $S_{\vec{n}}|\psi_1\rangle$, so we can just plug in the answer from there, and similarly the bottom entry of $S_{\vec{n}}|\psi_2\rangle$ is the top entry of $S_{\vec{n}}|\psi_1\rangle$.

The eigenvalue of $S_{\vec{n}}^2$ is $+\hbar^2/4$.